

Final Report : Food Delivery Time Prediction

1. Project Summary

This project focused on analyzing food delivery operations using Machine Learning techniques. Two models were developed:

- **Linear Regression** to predict exact delivery time.
- **Logistic Regression** to classify deliveries as Fast or Delayed.

The project included data preprocessing, exploratory data analysis (EDA), feature engineering, model training, evaluation, and business recommendations.

2. Dataset Description & Preprocessing

The dataset contained **200 records** and **15 features** related to food delivery services.

Important Features

- Distance
- Traffic Conditions
- Weather Conditions
- Vehicle Type
- Delivery Person Experience
- Order Priority
- Delivery Time

Preprocessing Steps

Checked for missing values with `df.isnull().sum()` and there were no missing data.

- Label Encoding was used to convert categorical variables to numerical values.
- Normalized numerical features like Distance and Order Cost using `StandardScaler`.
- Added a new feature `Rush_Hour` based on the order time to capture high traffic periods.

3. Model Evaluation & Comparison

Linear Regression

Used to predict exact delivery time.

Metric	Value
MSE	1.028
MAE	0.851
R ² Score	0.015

Observation:

The model showed weak performance because most features had low correlation with delivery time.

Logistic Regression

Used to classify deliveries as Fast or Delayed.

Metric	Value
Accuracy	62.5%
Precision	68.75%
Recall	52.38%
F1 Score	59.46%

Observation:

The model performed moderately well and gave better results compared to Linear Regression.

Comparison

Aspect	Linear Regression	Logistic Regression
Objective	Predict exact delivery time	Predict fast/delayed delivery
Performance	Weak	Moderate
Main Metric	R ² Score	Accuracy
Best Result	R ² = 0.015	Accuracy = 62.5%
Practical Usefulness	Limited	Better

Conclusion:

Logistic Regression was more effective for this dataset because classification was easier than predicting exact delivery time.

4. Actionable Insights & Recommendations

The following improvements are suggested based on the analysis:

a. Path Optimization

Minimize delivery delays with optimized routes.

b. Add Staff During Rush Hours

Use more delivery people during busy times.

c. Training for Delivery Staff

Improve navigation and time management abilities of delivery agents.

d. Integration of Real-Time Traffic

Use live traffic information to route around traffic and increase delivery productivity.

5. Final Conclusion

This project demonstrated the end to end machine learning workflow for food delivery prediction. Linear Regression did not perform as well, but Logistic Regression was able to classify deliveries as fast or delayed with moderate success. The project underscored the importance of preprocessing, feature engineering and model evaluation in improving prediction accuracy and operational decision-making.